The Architecture of Lifting and Extension

A Categorical and Topological Guide to Projective and Injective Objects

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Abstract

When first learning homological algebra in an abelian category $\mathcal{A}$, it is easy to get lost in a maze of flipped arrows: which of $\mathrm{Hom}_{\mathcal{A}}(A, -)$ and $\mathrm{Hom}_{\mathcal{A}}(-, A)$ is left exact, which object is projective, and which is injective? This essay resolves that confusion permanently by looking at the underlying categorical and topological architecture. We begin by showing how the Yoneda philosophy and the opposite category $\mathcal{A}^{\mathrm{op}}$ eliminate arrow-flipping confusion. We then take a deep dive into algebraic topology—specifically lifting problems (covering spaces and fibrations) and extension problems (Tietze extension and cofibrations)—to see how Daniel Quillen unified both concepts into a single commutative square. Finally, we bring this geometric intuition back to abelian categories to illuminate projective resolutions, chain homotopies, divisible groups, and sheaves.

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1 Deconstructing the Variance Trap: Why Hom is Always Left Exact

1.1 The Source of the Mental Friction

Suppose we are working in an abelian category $\mathcal{A}$. Every object $A \in \mathcal{A}$ gives us two fundamental ways to probe the category by landing in the category Ab of abelian groups:

1. The **covariant** Hom-functor $\mathrm{Hom}_{\mathcal{A}}(A, -) : \mathcal{A} \to \mathsf{Ab}$, which sends an object X to the group of morphisms *out of A and into X*.
2. The **contravariant** Hom-functor $\mathrm{Hom}_{\mathcal{A}}(-, A) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$, which sends an object X to the group of morphisms *out of X and into A*.

A classic trap when learning homological algebra is to think: *"Since* $\mathrm{Hom}(A, -)$ *is covariant and* $\mathrm{Hom}(-, A)$ *is contravariant, shouldn't one of them be left exact and the other be right exact?"*

The answer is a resounding **no**: *both* $\mathrm{Hom}(A, -)$ and $\mathrm{Hom}(-, A)$ are **left exact** (Proposition 2.2.13). Neither one is right exact in general.

Why does our intuition try to trick us here? Because if you think about exactness by visually looking at the "left side" and "right side" of a short exact sequence

$$0 \longrightarrow X' \xrightarrow{f} X \xrightarrow{g} X'' \longrightarrow 0$$

in $\mathcal{A}$, a contravariant functor reverses the arrows, which makes the right side of the sequence in $\mathcal{A}$ trade places with the left side of the sequence in Ab. Trying to track which side of the page is "preserved" while arrows are flipping in mid-air is a recipe for vertigo.

To cure this vertigo, we must stop thinking about "left and right sides of a page" and start thinking structurally about **limits and colimits**.

1.2 Limits, Colimits, and the $\mathcal{A}^{\mathrm{op}}$ Dictionary

Left exactness and right exactness are defined for a **covariant** additive functor $F : \mathcal{C} \to \mathcal{B}$ between abelian categories:

- F is **left exact** if it preserves **kernels** (or equivalently, all finite limits).
- F is **right exact** if it preserves **cokernels** (or equivalently, all finite colimits).

Notice why the words "left" and "right" were chosen in the first place: in any short exact sequence $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$, the object X' on the left is canonically the **kernel** of g ($X' \simeq \ker g$, which is a limit: the equalizer of g and 0), while the object X'' on the right is canonically the **cokernel** of f ($X'' \simeq \operatorname{coker} f$, which is a colimit: the coequalizer of f and 0).

Now, how do we apply this definition to the contravariant functor $\mathrm{Hom}_{\mathcal{A}}(-, A)$? A contravariant functor on $\mathcal{A}$ is literally a **covariant functor on the opposite category $\mathcal{A}^{\mathrm{op}}$**:

$$\mathrm{Hom}_{\mathcal{A}}(-, A) : \mathcal{A}^{\mathrm{op}} \longrightarrow \mathsf{Ab}.$$

Since $\mathcal{A}$ is an abelian category, $\mathcal{A}^{\mathrm{op}}$ is also an abelian category. Therefore, saying that $\mathrm{Hom}_{\mathcal{A}}(-, A) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ is **left exact** means one thing and one thing only: **as a covariant functor out of $\mathcal{A}^{\mathrm{op}}$, it preserves limits (kernels)**.

Let us write this out explicitly:

1. Suppose $\alpha : I \to \mathcal{A}$ is a small diagram in $\mathcal{A}$. By the very definition of a limit cone, giving a morphism from A into $\lim_I \alpha$ is identical to giving a compatible family of morphisms from A into each $\alpha(i)$. Thus, $\mathrm{Hom}_{\mathcal{A}}(A, -)$ pulls limits out:

$$\mathrm{Hom}_{\mathcal{A}}\left(A, \lim_{\mathcal{A}} \alpha\right) \simeq \lim_{\mathsf{Ab}} \mathrm{Hom}_{\mathcal{A}}(A, \alpha(i)).$$

2. Now suppose $\beta : I \to \mathcal{A}^{\mathrm{op}}$ is a diagram in $\mathcal{A}^{\mathrm{op}}$. What is the limit of β computed inside $\mathcal{A}^{\mathrm{op}}$? Because $\mathcal{A}^{\mathrm{op}}$ is obtained from $\mathcal{A}$ by reversing all arrows, a terminal cone in $\mathcal{A}^{\mathrm{op}}$ is literally an initial cocone in $\mathcal{A}$. That is:

$$\lim_{\mathcal{A}^{\mathrm{op}}} \beta = \mathrm{colim}_{\mathcal{A}}\, \beta.$$

   When we apply $\mathrm{Hom}_{\mathcal{A}}(-, A)$ to this object, the universal property of the colimit in $\mathcal{A}$ (that a map out of a colimit into A is a compatible family of maps out of each $\beta(i)$ into A) tells us:

$$\mathrm{Hom}_{\mathcal{A}}\left(\lim_{\mathcal{A}^{\mathrm{op}}} \beta, A\right) = \mathrm{Hom}_{\mathcal{A}}\left(\mathrm{colim}_{\mathcal{A}}\, \beta, A\right) \simeq \lim_{\mathsf{Ab}} \mathrm{Hom}_{\mathcal{A}}(\beta(i), A).$$

Look at the right-hand side of both equations: **in both cases, the operation in Ab is $\lim_{\mathsf{Ab}}$!**

- $\mathrm{Hom}_{\mathcal{A}}(A, -)$ takes a **limit (kernel) in $\mathcal{A}$** and turns it into a **limit (kernel) in Ab**.
- $\mathrm{Hom}_{\mathcal{A}}(-, A)$ takes a **colimit (cokernel) in $\mathcal{A}$**—which is a **limit (kernel) in $\mathcal{A}^{\mathrm{op}}$**—and turns it into a **limit (kernel) in Ab**.

Topological Intuition 1.1 (Why Ab Always Sees a Kernel). Even without opposite categories, you can see why both functors produce kernels in Ab just by looking at what a kernel is in Ab. In Ab, the kernel of any group homomorphism Φ is the subgroup of elements that Φ sends to 0.

- By definition of $\ker(g : B \to C)$ in $\mathcal{A}$, morphisms $\varphi : A \to \ker g$ are precisely morphisms $\varphi : A \to B$ satisfying $g \circ \varphi = 0$. In Ab, the set of φ solving $g_*(\varphi) = g \circ \varphi = 0$ is a **kernel**: $\ker(g_*)$.
- By definition of $\mathrm{coker}(f : A \to B)$ in $\mathcal{A}$, morphisms $\psi : \mathrm{coker}\, f \to I$ are precisely morphisms $\psi : B \to I$ satisfying $\psi \circ f = 0$. In Ab, the set of ψ solving $f^*(\psi) = \psi \circ f = 0$ is *also* a **kernel**: $\ker(f^*)$.

Because "composing to zero" always defines a kernel in Ab, both Hom functors always manufacture kernels in Ab, and therefore both are **left exact**.

1.3 The Strictness Rescue: Exactness at the Middle Term

Let us make sure one crucial technical detail is completely solid. Suppose we have a short exact sequence in $\mathcal{A}$:

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0.$$

This means $A \simeq \ker g$ and $C \simeq \mathrm{coker}\, f$, so we may write the sequence canonically as:

$$0 \longrightarrow \ker g \xhookrightarrow{\iota} B \xrightarrow{\pi} \mathrm{coker}\, f \longrightarrow 0.$$

When we apply $\mathrm{Hom}_{\mathcal{A}}(P, -)$ for any object $P \in \mathcal{A}$, we obtain a sequence of abelian groups:

$$0 \longrightarrow \mathrm{Hom}(P, \ker g) \xrightarrow{\iota_*} \mathrm{Hom}(P, B) \xrightarrow{\pi_*} \mathrm{Hom}(P, \mathrm{coker}\, f).$$

Recall that a complex $X' \xrightarrow{\alpha} X \xrightarrow{\beta} X''$ in an abelian category is exact at X if and only if $\operatorname{im}(\alpha) = \ker(\beta)$ as subobjects of X, where $\operatorname{im}(\alpha) := \ker(X \to \operatorname{coker}\alpha)$.

If we try to verify exactness of our Hom-sequence at the middle term $\operatorname{Hom}(P, B)$ by naively plugging in the definition $\operatorname{im}(\iota_*) = \ker(\operatorname{Hom}(P, B) \to \operatorname{coker}(\iota_*))$, we hit a psychological wall: since $\operatorname{Hom}(P, -)$ does not preserve cokernels in general, we have no obvious description of $\operatorname{coker}(\iota_*)$!

How does the structure of an abelian category rescue us here? Through **strictness**:

1. First, because $\iota : \ker g \hookrightarrow B$ is a monomorphism in $\mathcal{A}$, for any $h : P \to \ker g$ we have $\iota_*(h) = \iota \circ h = 0 \implies h = 0$. Thus $\ker(\iota_*) = 0$, which means $\iota_* : \operatorname{Hom}(P, \ker g) \hookrightarrow \operatorname{Hom}(P, B)$ is a **monomorphism** in Ab (giving exactness at $\operatorname{Hom}(P, \ker g)$).

2. Second, what is the image $\operatorname{im}(m)$ of *any* monomorphism $m : X \hookrightarrow Y$ in an abelian category? Since m is mono, $\ker(m) = 0$, so its coimage is $\operatorname{coim}(m) := \operatorname{coker}(\ker m \to X) = X$. Because every morphism in an abelian category is **strict** ($\operatorname{coim} m \xrightarrow{\sim} \operatorname{im} m$), the subobject $\operatorname{im}(m) \hookrightarrow Y$ is canonically isomorphic to $X \overset{m}{\hookrightarrow} Y$ itself!

3. Applying this to the monomorphism ι_*, we get its image immediately *without ever computing* $\operatorname{coker}(\iota_*)$:
$$\operatorname{im}(\iota_*) = \big(\operatorname{Hom}(P, \ker g) \overset{\iota_*}{\hookrightarrow} \operatorname{Hom}(P, B)\big).$$

4. Finally, because $\operatorname{Hom}(P, -)$ preserves kernels, what is $\ker(\pi_*)$? It is:
$$\ker(\pi_*) = \operatorname{Hom}(P, \ker \pi) = \big(\operatorname{Hom}(P, \ker g) \overset{\iota_*}{\hookrightarrow} \operatorname{Hom}(P, B)\big).$$

Comparing steps 3 and 4, we see that $\operatorname{im}(\iota_*) = \ker(\pi_*)$ on the nose! In fact, saying $0 \to X' \xrightarrow{\alpha} X \xrightarrow{\beta} X''$ is exact at both X' and X is completely equivalent to saying that $\alpha : X' \to X$ is the kernel of $\beta : X \to X''$. Because $\operatorname{Hom}(P, -)$ preserves kernels, it preserves exactness at the first two spots in one single stroke.

2 Two Topological Problems That Turn Out to Be One

Now that we know $\operatorname{Hom}(P, -)$ and $\operatorname{Hom}(-, I)$ automatically preserve the left-hand side of exactness (kernels), we know that exactness can only ever fail at the right-hand end of the output sequence in Ab—namely, the final map in Ab might fail to be an epimorphism (surjection):

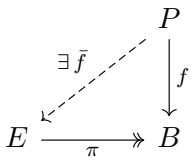
- An object $P \in \mathcal{A}$ is **projective** if $\operatorname{Hom}_{\mathcal{A}}(P, -)$ is exact, i.e., for every epimorphism $\pi : B \twoheadrightarrow C$, the induced map $\pi_* : \operatorname{Hom}(P, B) \to \operatorname{Hom}(P, C)$ is surjective.
- An object $I \in \mathcal{A}$ is **injective** if $\operatorname{Hom}_{\mathcal{A}}(-, I)$ is exact, i.e., for every monomorphism $i : A \hookrightarrow B$, the induced map $i^* : \operatorname{Hom}(B, I) \to \operatorname{Hom}(A, I)$ is surjective.

To build deep, unshakeable intuition for why these two definitions look the way they do, let us leave algebra for a moment and step into **topology**. In topology, geometers constantly face two fundamental problems: **Lifting** and **Extension**.

2.1 Problem 1: The Lifting Problem (Projective Flavor)

Suppose you have a projection map $\pi : E \twoheadrightarrow B$ from a "total space" E down to a "base space" B (think of a covering space $\tilde{X} \to X$, or a fiber bundle projecting a 3D object onto its 2D shadow). You also have a test space P and a map $f : P \to B$ landing in the shadow B.

The Lifting Question: Can we factor f through the total space E? That is, does there exist a continuous map $\bar{f} : P \to E$ (a *lift* of f) such that $\pi \circ \bar{f} = f$?



Notice the geometry here:

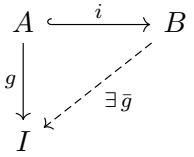
- The given map $\pi : E \twoheadrightarrow B$ is a **projection** (surjection/epimorphism).
- Our fixed object P is the **domain** (the source of the maps $P \to B$ and $P \to E$), so in $\mathrm{Hom}(-,-)$ it sits in the **first slot**: $\mathrm{Hom}(P,E) \xrightarrow{\pi_*} \mathrm{Hom}(P,B)$.
- What prevents a lift $\bar{f}$ from existing? **Internal relations or loops inside P!** For example, if $E = \mathbb{R} \xrightarrow{\pi} S^1 = B$ is the universal covering helix of the circle, and we take $P = S^1$ with $f = \mathrm{id}_{S^1} : S^1 \to S^1$, we *cannot* lift f to $\mathbb{R}$. Why? Because as you walk around $P = S^1$, you return to your starting point ($1 \cdot x = x$), whereas lifting that path upstairs to $\mathbb{R}$ takes you from 0 to $1 \neq 0$. The non-trivial loop in P ($\pi_1(S^1) = \mathbb{Z}$) is an obstruction to lifting!
- Conversely, if P is **simply connected** ($\pi_1(P) = 0$, or contractible), then P has no internal loop obstructions, so *every* map $f : P \to B$ lifts across *every* covering space $\pi : E \to B$!

In homological algebra, a **projective object** P is the exact algebraic analogue of a simply connected/unobstructed domain: it is an object so free of internal relations that any map from P into a quotient C can be **lifted** across any epimorphism $\pi : B \twoheadrightarrow C$.

2.2 Problem 2: The Extension Problem (Injective Flavor)

Now consider the dual geometric problem. Suppose you have a subspace A sitting inside a larger ambient space B via an inclusion map $i : A \hookrightarrow B$ (think of the boundary circle S^1 sitting inside the disk D^2, or a closed subset $A \subset B$). You have a map $g : A \to I$ defined *only on the subspace* A, taking values in some target space I.

The Extension Question: Can we extend the domain of g from the small subspace A to the entire ambient space B? That is, does there exist a map $\bar{g} : B \to I$ (an *extension* of g) such that $\bar{g}|_A = \bar{g} \circ i = g$?



Notice how every single piece of the geometry has reversed:

- The given map $i : A \hookrightarrow B$ is an **injection** (inclusion/monomorphism).
- Our fixed object I is the **target** (the codomain receiving the maps $A \to I$ and $B \to I$), so in $\mathrm{Hom}(-,-)$ it sits in the **second slot**: $\mathrm{Hom}(B,I) \xrightarrow{i^*} \mathrm{Hom}(A,I)$.
- What prevents an extension $\bar{g}$ from existing? **Rigidity or holes in the target I!** For example, if $i : S^1 \hookrightarrow D^2$ is the boundary of a disk, and our target is $I = S^1$ with $g = \mathrm{id}_{S^1}$, we *cannot* extend g to $\bar{g} : D^2 \to S^1$, because the hole in the target S^1 prevents us from filling in the center of the disk continuously.

- Conversely, what if we change the target space I to the real line $\mathbb{R}$ (or any Euclidean space $\mathbb{R}^n$)? By the **Tietze Extension Theorem**, if $A \hookrightarrow B$ is any closed subspace of a normal space B, *every* continuous map $g : A \to \mathbb{R}$ extends to a continuous map $\bar{g} : B \to \mathbb{R}$! Why? Because $\mathbb{R}$ is convex and divisible: you can average, scale, and interpolate values in $\mathbb{R}$ across the ambient space B without ever getting trapped by a hole.

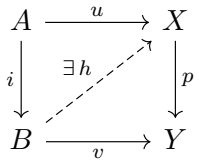
In homological algebra, an **injective object** I is the exact algebraic analogue of the Tietze target $\mathbb{R}$: it is a target object so "divisible" and accommodating that any morphism defined on a subobject $A \hookrightarrow B$ can be **extended** to the entire object $B \to I$.

2.3 The Categorical Miracle: Fusing Lifting and Extension into One Square

At first glance, the Lifting Problem (a triangle with an epimorphism $\pi : E \twoheadrightarrow B$ at the bottom) and the Extension Problem (a triangle with a monomorphism $i : A \hookrightarrow B$ at the top) look like two different diagrams.

In 1967, Daniel Quillen realized that if you put a map $i : A \to B$ on the **left** and a map $p : X \to Y$ on the **right** of a commutative square, **Lifting and Extension become the exact same problem**:

Definition 2.1 (The Lifting Square / Weak Factorization System)**.** Let $i : A \to B$ and $p : X \to Y$ be morphisms in any category $\mathcal{C}$. We say that i has the **Left Lifting Property (LLP)** with respect to p, and p has the **Right Lifting Property (RLP)** with respect to i—written $i \pitchfork p$—if for every commutative square of solid arrows



there exists a diagonal morphism $h : B \to X$ making both triangles commute ($h \circ i = u$ and $p \circ h = v$).

Look closely at what the diagonal arrow $h : B \to X$ is doing in that square:

- Viewed relative to the **upper-left triangle** ($h \circ i = u$), $h : B \to X$ is an **extension** of $u : A \to X$ along $i : A \to B$!
- Viewed relative to the **lower-right triangle** ($p \circ h = v$), $h : B \to X$ is a **lift** of $v : B \to Y$ across $p : X \to Y$!

In other words, a diagonal filler h in the square is *simultaneously* an extension of the top map and a lift of the bottom map!

3 Fibrations and Cofibrations for the Working Algebraist

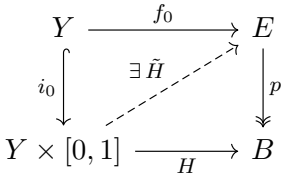
The lifting square $i \pitchfork p$ naturally separates into two complementary perspectives:

- When you fix a class of inclusions i on the **left**, the maps p on the **right** that satisfy $i \pitchfork p$ are called **fibrations**.
- When you fix a class of projections p on the **right**, the maps i on the **left** that satisfy $i \pitchfork p$ are called **cofibrations**.

3.1 Fibrations: The Right-Hand Side of the Square

In topology, what makes a projection $p : E \to B$ (like a covering space or a fiber bundle) "well-behaved"? If you have a continuous map $f_0 : Y \to E$ upstairs, and you continuously deform its shadow $p \circ f_0 : Y \to B$ downstairs via a homotopy $H : Y \times [0,1] \to B$, you should be able to lift that entire deformation upstairs to a homotopy $\tilde{H} : Y \times [0,1] \to E$ without getting stuck on a "cliff edge" in E.

Let $I = [0,1]$ be the unit interval, and let $i_0 : Y \hookrightarrow Y \times [0,1]$ be the inclusion of the bottom of the cylinder, $y \mapsto (y, 0)$. Then the data of "a map $f_0 : Y \to E$ upstairs and a homotopy $H : Y \times [0,1] \to B$ downstairs starting at $p \circ f_0$" is literally a commutative square:



Definition 3.1 (Hurewicz Fibration)**.** A continuous map $p : E \to B$ is a **fibration** if it has the **Homotopy Lifting Property (HLP)** with respect to all spaces Y—meaning p has the **Right Lifting Property** ($i_0 \pitchfork p$) with respect to all cylinder inclusions $i_0 : Y \hookrightarrow Y \times [0,1]$.

Every covering space $\tilde{X} \to X$ is a fibration (with discrete fibers), and every vector bundle or fiber bundle is a fibration. Notice that in the lifting square, the fibration $p : E \to B$ always sits on the **right vertical edge**.

3.2 Cofibrations: The Left-Hand Side of the Square

Now let us dualize. What makes a subspace inclusion $i : A \hookrightarrow B$ in topology "well-behaved" (like the boundary of a disk $S^{n-1} \hookrightarrow D^n$, or a subcomplex of a CW-complex)?

Suppose you have a continuous map $g_0 : B \to X$ defined on the whole space B, and on the subspace A you continuously deform $g_0|_A$ via a homotopy $G : A \times [0,1] \to X$. If A sits inside B nicely (with a "collar neighborhood" around it so you have room to blend the deformation continuously into the rest of B), then you can **extend** that homotopy to a deformation $\tilde{G} : B \times [0,1] \to X$ on the entire space B. This is called the **Homotopy Extension Property (HEP)**.

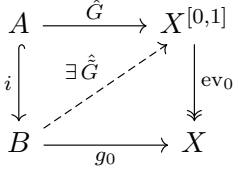
How is the Homotopy Extension Property a lifting square? By the **exponential adjunction** in topology:

$$\mathrm{Hom}_{\mathsf{Top}}(A \times [0,1], X) \simeq \mathrm{Hom}_{\mathsf{Top}}(A, X^{[0,1]}),$$

where $X^{[0,1]} = \mathrm{Hom}_{\mathsf{Top}}([0,1], X)$ is the **path space** of X (the space of all continuous paths $\gamma : [0,1] \to X$), equipped with the evaluation map at time 0:

$$\mathrm{ev}_0 : X^{[0,1]} \twoheadrightarrow X, \qquad \gamma \longmapsto \gamma(0).$$

Under this adjunction, a homotopy $G : A \times [0,1] \to X$ is simply a map $\hat{G} : A \to X^{[0,1]}$ sending each point $a \in A$ to its trajectory path in X! Saying that the homotopy starts at $g_0|_A$ at time $t = 0$ is saying that $\mathrm{ev}_0 \circ \hat{G} = g_0 \circ i$. And extending the homotopy to all of B means finding a map $\hat{\tilde{G}} : B \to X^{[0,1]}$ making the square below commute:



Definition 3.2 (Cofibration)**.** A continuous map $i : A \hookrightarrow B$ is a **cofibration** if it satisfies the **Homotopy Extension Property (HEP)** with respect to all target spaces X—meaning i has the **Left Lifting Property** ($i \pitchfork \mathrm{ev}_0$) with respect to all path-space fibrations $\mathrm{ev}_0 : X^{[0,1]} \twoheadrightarrow X$.

Look at the symmetry we have uncovered:

- **Cofibrations** ($i : A \hookrightarrow B$) live on the **left vertical edge** of the square and govern **extension problems**.
- **Fibrations** ($p : E \twoheadrightarrow B$) live on the **right vertical edge** of the square and govern **lifting problems**.

4 Back to Abelian Categories: Projectives and Injectives in 3D

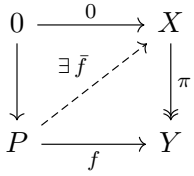
4.1 Projectives Are Cofibrant; Injectives Are Fibrant

Now let us bring Quillen's lifting square directly back to an abelian category $\mathcal{A}$.

In an abelian category $\mathcal{A}$, the two natural classes of "nice inclusions" and "nice projections" are the **monomorphisms** ($i : A \hookrightarrow B$) and the **epimorphisms** ($\pi : X \twoheadrightarrow Y$). Furthermore, $\mathcal{A}$ has a zero object 0, which is both initial (every object P has a unique map $0 \to P$) and terminal (every object I has a unique map $I \to 0$).

Watch what happens when we plug $0 \to P$ into the **left** edge of the lifting square, and $I \to 0$ into the **right** edge of the lifting square:

1. **Projective Objects ($0 \to P$ on the Left):** Put $0 \to P$ on the left and an arbitrary epimorphism $\pi : X \twoheadrightarrow Y$ on the right:

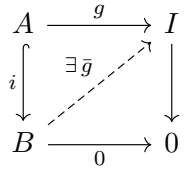


   Because 0 is the zero object, the top horizontal arrow $0 \to X$ is forced to be the zero morphism, and the upper-left triangle commutes automatically for any $\bar{f} : P \to X$. Thus, the existence of a diagonal filler $\bar{f} : P \to X$ is literally the projective lifting property $\pi \circ \bar{f} = f$!

$$\boxed{P \text{ is projective } \iff (0 \to P) \pitchfork (\text{all epimorphisms } \pi : X \twoheadrightarrow Y).}$$

   In Quillen's terminology, an object P whose initial map $0 \to P$ has the Left Lifting Property (behaving like a cofibration) is called **cofibrant**. *Projective objects are cofibrant objects.*

2. **Injective Objects ($I \to 0$ on the Right):** Put an arbitrary monomorphism $i : A \hookrightarrow B$ on the left and $I \to 0$ on the right:



Because 0 is the zero object, the bottom horizontal arrow $B \to 0$ is forced to be 0, and the lower-right triangle commutes automatically for any $\bar{g} : B \to I$. Thus, the existence of a diagonal filler $\bar{g} : B \to I$ is literally the injective extension property $\bar{g} \circ i = g$!

$$\boxed{I \text{ is injective } \iff (\text{all monomorphisms } i : A \hookrightarrow B) \pitchfork (I \to 0).}$$

In Quillen's terminology, an object I whose terminal map $I \to 0$ has the Right Lifting Property (behaving like a fibration) is called **fibrant**. *Injective objects are fibrant objects.*

4.2 Visualizing Proofs via the Lifting Square: (Co)products and Adjunctions

Once you view projective and injective objects through the lifting square, standard results in homological algebra become visually self-evident.

1. Why Coproducts of Projectives are Projective, and Products of Injectives are Injective

Suppose $\{P_\alpha\}$ is a family of objects whose coproduct $\bigoplus_\alpha P_\alpha$ exists in $\mathcal{A}$. Why is $\bigoplus_\alpha P_\alpha$ projective if and only if each P_α is projective? Look at the left side of the lifting square: by the universal property of the coproduct (initial cocone), specifying a morphism out of $\bigoplus_\alpha P_\alpha$ into X is identical to specifying an independent morphism $P_\alpha \to X$ on each summand. Thus, you can lift $\bigoplus_\alpha P_\alpha \to Y$ across $\pi : X \twoheadrightarrow Y$ if and only if you can lift each piece $P_\alpha \to Y$ across π independently!

Dually, why is a **product** $\prod_\alpha I_\alpha$ injective if and only if each I_α is injective? Look at the right side of the lifting square: by the universal property of the product (terminal cone), specifying an extension $B \to \prod_\alpha I_\alpha$ into a product is identical to specifying an independent extension $B \to I_\alpha$ into each factor!

2. How Adjunctions Transport Projectives and Injectives

Consider an adjoint pair of additive functors between abelian categories:

$$F : \mathcal{A} \rightleftarrows \mathcal{B} : G, \qquad \mathrm{Hom}_{\mathcal{B}}(F(-), -) \simeq \mathrm{Hom}_{\mathcal{A}}(-, G(-)).$$

A classic theorem states:

- If F is exact, then its right adjoint G preserves injectives.
- If G is exact, then its left adjoint F preserves projectives.

Why is this true structurally? Because in *any* adjunction $F \dashv G$, the bijection $\mathrm{Hom}_{\mathcal{B}}(F(A), X) \simeq \mathrm{Hom}_{\mathcal{A}}(A, G(X))$ allows you to **transpose a lifting square from $\mathcal{B}$ to $\mathcal{A}$**:

$$\begin{array}{ccc} F(A) & \xrightarrow{u} & X \\ {\scriptstyle F(i)}\downarrow & \overset{\exists\, h}{\nearrow} & \downarrow{\scriptstyle p} \\ F(B) & \xrightarrow[v]{} & Y \end{array} \text{ in } \mathcal{B} \qquad \iff \qquad \begin{array}{ccc} A & \xrightarrow{\hat{u}} & G(X) \\ {\scriptstyle i}\downarrow & \overset{\exists\, \hat{h}}{\nearrow} & \downarrow{\scriptstyle G(p)} \\ B & \xrightarrow[\hat{v}]{} & G(Y) \end{array} \text{ in } \mathcal{A}.$$

In symbols:

$$F(i) \pitchfork p \text{ in } \mathcal{B} \quad \iff \quad i \pitchfork G(p) \text{ in } \mathcal{A}.$$

Look at what this single equivalence gives you immediately:

- Take $i = (0 \to P)$ for a projective $P \in \mathcal{A}$. Then $F(0 \to P) = (0 \to F(P))$ lifts against an epimorphism $p : X \twoheadrightarrow Y$ in $\mathcal{B}$ if and only if $(0 \to P)$ lifts against $G(p) : G(X) \to G(Y)$ in $\mathcal{A}$. If G is exact, G preserves epimorphisms, so $G(p)$ is epi in $\mathcal{A}$, hence the lift $\hat{h}$ exists because P is projective, which transposes back to h in $\mathcal{B}$! Thus $F(P)$ is projective.
- Dually, take $p = (J \to 0)$ for an injective $J \in \mathcal{B}$. Then $G(J \to 0) = (G(J) \to 0)$ lifts on the right against a monomorphism $i : A \hookrightarrow B$ in $\mathcal{A}$ if and only if $(J \to 0)$ lifts on the right against $F(i) : F(A) \to F(B)$ in $\mathcal{B}$. If F is exact, F preserves monomorphisms, so $F(i)$ is mono in $\mathcal{B}$, hence the extension h exists because J is injective, which transposes to $\hat{h}$ in $\mathcal{A}$! Thus $G(J)$ is injective.

4.3 What About Free Objects and Generators?

In R-Mod, we build projective modules by starting with **free modules** $R^{\oplus S}$ (which are projective, and every projective module is a direct summand of a free module). Why can't we talk about "free objects" in a general abelian category $\mathcal{A}$?

Because "freeness" is not an internal property of a single category—it is a property of a **Free-Forgetful adjunction**:

$$\text{Free} : \mathsf{Set} \rightleftarrows R\text{-}\mathsf{Mod} : U.$$

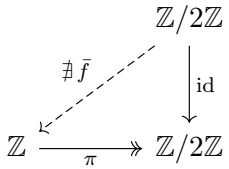
By our transposition rule above, since the forgetful functor $U : R\text{-}\mathsf{Mod} \to \mathsf{Set}$ sends epimorphisms of modules to surjective functions of sets, and in Set *every* set S is projective (thanks to the Axiom of Choice, every surjective function of sets has a section!), the left adjoint $\text{Free}(S) = R^{\oplus S}$ automatically sends any set S to a projective R-module!

In an abstract abelian category $\mathcal{A}$ (like the category of sheaves on a topological space), there is no canonical forgetful functor to Set. How does category theory replace the role of the "free module of rank 1" ($R \in R\text{-}\mathsf{Mod}$)? With the concept of a **generator**: an object $S \in \mathcal{A}$ such that every object $X \in \mathcal{A}$ admits an epimorphism $\bigoplus_{i \in I} S \twoheadrightarrow X$ from a coproduct of copies of S. If $\mathcal{A}$ happens to have a *projective* generator S (like R in R-Mod), then coproducts $\bigoplus_{i \in I} S$ play the exact role of free objects, guaranteeing that $\mathcal{A}$ has enough projectives!

5 Concrete Topological and Algebraic Case Studies

5.1 Case Study 1: Torsion vs. Fundamental Group ($\mathbb{Z}/2\mathbb{Z}$ is Not Projective)

Let us look at the diagram of the natural quotient epimorphism $\pi : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$:



Why is there no lift $\bar{f} : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$? Because inside the domain $\mathbb{Z}/2\mathbb{Z}$, the generator $\bar{1}$ satisfies the **torsion relation** $2 \cdot \bar{1} = \bar{0}$. Any group homomorphism $\bar{f} : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ must preserve this relation:

$$2 \cdot \bar{f}(\bar{1}) = \bar{f}(2 \cdot \bar{1}) = \bar{f}(\bar{0}) = 0 \in \mathbb{Z}.$$

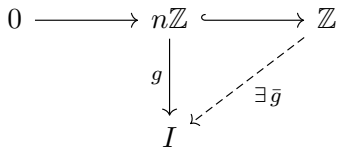
Because $\mathbb{Z}$ is torsion-free, the only element in $\mathbb{Z}$ satisfying $2x = 0$ is $x = 0$. Thus $\bar{f} = 0$ is the *only* homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$, which projects down to $0 \neq \text{id}_{\mathbb{Z}/2\mathbb{Z}}$.

Compare this to the covering space $\pi : S^2 \twoheadrightarrow \mathbb{R}P^2$ in topology! The real projective plane $\mathbb{R}P^2$ has fundamental group $\pi_1(\mathbb{R}P^2) = \mathbb{Z}/2\mathbb{Z}$ (a loop γ that is non-trivial, but walking around it *twice* gives a contractible loop $2\gamma = 0$). You cannot lift $\mathrm{id} : \mathbb{R}P^2 \to \mathbb{R}P^2$ to the universal cover S^2 because S^2 is simply connected ($\pi_1(S^2) = 0$), so the 2-torsion loop γ has nowhere to go! **Algebraic torsion in a module is the exact counterpart of a non-trivial loop in π_1.**

5.2 Case Study 2: Divisibility vs. Convexity (Baer's Criterion as Tietze Extension)

An abelian group $I \in \mathsf{Ab}$ is injective if and only if it is **divisible** ($nI = I$ for all $n \neq 0$, such as $\mathbb{Q}$ or $\mathbb{Q}/\mathbb{Z}$).

Why does "injective = divisible" in $\mathbb{Z}$-Mod? By **Baer's Criterion**, to check whether a module I is injective, it suffices to test extensions along inclusions of ideals $\mathfrak{a} \hookrightarrow R$. For $R = \mathbb{Z}$, every non-zero ideal is of the form $n\mathbb{Z} \hookrightarrow \mathbb{Z}$:



A homomorphism $g : n\mathbb{Z} \to I$ is completely determined by choosing an element $y = g(n) \in I$. To extend g to a homomorphism $\bar{g} : \mathbb{Z} \to I$, we must choose where to send the generator $1 \in \mathbb{Z}$, say $\bar{g}(1) = x \in I$. Compatibility ($\bar{g}(n) = g(n)$) requires:

$$n \cdot x = y \quad \text{in } I.$$

Therefore, an extension $\bar{g}$ exists for every $g : n\mathbb{Z} \to I$ if and only if *every element $y \in I$ can be divided by n* ($x = \frac{1}{n}y \in I$)!

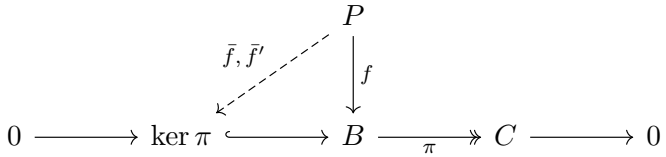
Notice how clean the geometric analogy is: in Tietze's Extension Theorem, extending a continuous function from the boundary $\{0, n\} \hookrightarrow [0, n]$ to the interior points $1, 2, \ldots, n-1$ requires being able to *subdivide/interpolate* between $g(0)$ and $g(n)$ inside the target $\mathbb{R}$. In Ab, extending a homomorphism from the sublattice $n\mathbb{Z} \hookrightarrow \mathbb{Z}$ to the intermediate integer $1 \in \mathbb{Z}$ literally requires dividing $g(n)$ by n inside I!

5.3 Case Study 3: Non-Uniqueness of Lifts and Why Chain Homotopies Appear

In topology, if $\pi : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ is a pointed covering space, the fiber $\pi^{-1}(x_0)$ is **discrete**. Because the fiber is discrete, once you fix the basepoint $\tilde{x}_0$, the lift of any map from a simply connected space is **unique on the nose**.

What if $\pi : E \to B$ is a general fibration whose fibers $\pi^{-1}(b)$ are *not* discrete (for instance, the fibers are continuous spaces)? Then a lift $\bar{f} : P \to E$ is **no longer unique on the nose**—instead, any two lifts $\bar{f}_0, \bar{f}_1 : P \to E$ can be deformed into each other along the fiber, meaning they are **unique up to homotopy**!

Now look at what happens in an abelian category $\mathcal{A}$ when P is projective:



Is the lift $\bar{f} : P \to B$ unique? **No!** The "fiber" of π over 0 is $\ker \pi$, which is generally non-zero. If $\bar{f}$ and $\bar{f}'$ are two different lifts of f ($\pi\bar{f} = f = \pi\bar{f}'$), then their difference $\bar{f} - \bar{f}' : P \to B$ satisfies $\pi(\bar{f} - \bar{f}') = 0$, so by the universal property of the kernel it factors through a morphism into the fiber:

$$h_0 : P \longrightarrow \ker \pi \hookrightarrow B.$$

When we resolve an object X by a projective resolution $\cdots \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} X \to 0$ and lift a morphism $\varphi : X \to Y$ to a chain map $f_\bullet : P_\bullet \to J_\bullet$ between projective resolutions:

1. At degree 0, we use the projectivity of P_0 to lift $\varphi \circ \epsilon : P_0 \to Y$ across $J_0 \twoheadrightarrow Y$ to get $f_0 : P_0 \to J_0$.
2. Why isn't f_0 unique? Because someone else could choose another lift $g_0 : P_0 \to J_0$! The difference $f_0 - g_0$ lands in $\ker(J_0 \twoheadrightarrow Y) = \mathrm{im}(d_1^J : J_1 \to J_0)$.
3. Because P_0 is projective and $J_1 \twoheadrightarrow \mathrm{im}(d_1^J)$ is an epimorphism, we can lift the discrepancy $f_0 - g_0 : P_0 \to \mathrm{im}(d_1^J)$ one step to the left to get a diagonal map $h_0 : P_0 \to J_1$ satisfying $d_1^J h_0 = f_0 - g_0$!
4. Inductively, at every degree n, the ambiguity in choosing the projective lift $f_n : P_n \to J_n$ is absorbed by a diagonal lift $h_n : P_n \to J_{n+1}$ into the next syzygy:

$$f_n - g_n = d_{n+1}^J h_n + h_{n-1} d_n^P.$$

Moral: The reason chain homotopies exist in homological algebra is that projective lifts are not unique on the nose; the ambiguity between two lifts lives in the kernel $\ker d_n = \mathrm{im}\, d_{n+1}$, and lifting that ambiguity across $J_{n+1} \twoheadrightarrow \mathrm{im}\, d_{n+1}$ is literally a chain homotopy!

5.4 Case Study 4: Sheaves on a Topological Space ($\mathsf{Sh}(X)$)

For any topological space X, the category $\mathsf{Sh}(X)$ of sheaves of abelian groups on X is an abelian category that has **enough injectives**. However, on almost any connected space (like $X = \mathbb{R}$ or S^1), $\mathsf{Sh}(X)$ does **not** have enough projectives (in fact, the only projective sheaf on $\mathbb{R}$ is the zero sheaf 0).

Our topological lifting dictionary explains this dramatic asymmetry immediately:

- **Why Projective Sheaves Fail to Exist:** In $\mathsf{Sh}(X)$, a morphism of sheaves $\pi : \mathcal{F} \twoheadrightarrow \mathcal{G}$ is an epimorphism if and only if it is surjective *locally on stalks* (around every point $x \in X$, you can shrink to a tiny open neighborhood $U \ni x$ where a section lifts). However, globally over a large open set V, sections of $\mathcal{G}(V)$ do *not* need to lift to $\mathcal{F}(V)$ (for example, the exponential map of sheaves $\mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^\times$ is a sheaf epimorphism on $\mathbb{C}^\times$, even though $z \in \mathcal{O}_{\mathbb{C}}^\times(\mathbb{C}^\times)$ has no global logarithm on $\mathbb{C}^\times$!). Because you can always build sheaf epimorphisms that require passing to finer and finer open covers before a lift exists, no non-zero sheaf $\mathcal{P}$ on $\mathbb{R}$ can lift globally across *every* sheaf epimorphism!
- **Why Injective Sheaves Are Abundant:** To build an injective sheaf $\mathcal{I}$ on X, we go to the opposite extreme: at each individual point $x \in X$, pick an injective abelian group I_x (like $\mathbb{Q}$ or $\mathbb{Q}/\mathbb{Z}$), and form the **skyscraper sheaf** $x_*(I_x)$ supported at the single point x. Notice that evaluating a sheaf $\mathcal{F}$ at the stalk $\mathcal{F}_x$ is an **exact left adjoint** to the skyscraper functor $x_* : \mathsf{Ab} \to \mathsf{Sh}(X)$:

$$\mathrm{Hom}_{\mathsf{Sh}(X)}(\mathcal{F}, x_*(I_x)) \simeq \mathrm{Hom}_{\mathsf{Ab}}(\mathcal{F}_x, I_x).$$

By our adjunction transposition, because stalk-taking $\mathcal{F} \mapsto \mathcal{F}_x$ is exact, its right adjoint x_* sends the injective abelian group I_x to an **injective sheaf** $x_*(I_x) \in \mathsf{Sh}(X)$! And taking the **product** over all points $x \in X$ gives an injective sheaf $\mathcal{I} = \prod_{x \in X} x_*(I_x)$ into which our original sheaf embeds! Because sections of this product sheaf are just arbitrary, discontinuous point-by-point functions $U \to \prod_{x \in U} I_x$, extending a section from a smaller open set or subsheaf has zero topological glue holding it back.

Summary Cheat-Sheet: The Complete Structural Dictionary

Structural Feature	Projective Objects (P)	Injective Objects (I)
Functorial Definition	$\mathrm{Hom}_{\mathcal{A}}(P, -) : \mathcal{A} \to \mathsf{Ab}$ is exact.	$\mathrm{Hom}_{\mathcal{A}}(-, I) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ is exact.
Automatic Half (Left Exact)	Preserves kernels in $\mathcal{A}$ ($\ker_{\mathcal{A}} \mapsto \ker_{\mathsf{Ab}}$).	Turns cokernels in $\mathcal{A}$ (kernels in $\mathcal{A}^{\mathrm{op}}$) into kernels in Ab ($\mathrm{coker}_{\mathcal{A}} \mapsto \ker_{\mathsf{Ab}}$).
Extra Condition (Right Exact)	Preserves epimorphisms/cokernels: $\pi : B \twoheadrightarrow C \implies \pi_* : \mathrm{Hom}(P, B) \twoheadrightarrow \mathrm{Hom}(P, C)$.	Turns monomorphisms/kernels in $\mathcal{A}$ into epimorphisms in Ab: $i : A \hookrightarrow B \implies i^* : \mathrm{Hom}(B, I) \twoheadrightarrow \mathrm{Hom}(A, I)$.
Geometric Action	**Lifting** a map $P \to C$ upstairs across a projection $\pi : B \twoheadrightarrow C$.	**Extending** a map $A \to I$ outward from a subobject $i : A \hookrightarrow B$.
Quillen Lifting Square ($i \pitchfork p$)	$(0 \to P)$ on the **left** lifts against all epimorphisms on the right (**Cofibrant**).	$(I \to 0)$ on the **right** lifts against all monomorphisms on the left (**Fibrant**).
Topological Analogue	Simply connected / contractible domain Y ($\pi_1(Y) = 0$) lifting across a covering space or fibration $\tilde{X} \twoheadrightarrow X$.	Convex / contractible target $\mathbb{R}$ in Tietze Extension receiving extensions from closed subspaces $A \hookrightarrow B$.
Prototype in $\mathsf{Ab} = \mathbb{Z}$-Mod	**Free abelian groups** $\mathbb{Z}^{\oplus S}$ (no torsion relations $nx = 0$ obstructing generators).	**Divisible groups** $\mathbb{Q}, \mathbb{Q}/\mathbb{Z}$ (can always solve $nx = y$ to extend from $n\mathbb{Z} \hookrightarrow \mathbb{Z}$).
Closure Under (Co)limits	Closed under **coproducts** $\bigoplus_\alpha P_\alpha$ and direct summands.	Closed under **products** $\prod_\alpha I_\alpha$ and direct summands.
Preservation by Adjoints ($F \dashv G$)	Left adjoint F preserves projectives if right adjoint G is exact.	Right adjoint G preserves injectives if left adjoint F is exact.
Resolution Direction	Leftward (bounded above, $C^-(\mathcal{A})$): $\cdots \to P_1 \to P_0 \twoheadrightarrow X \to 0$.	Rightward (bounded below, $C^+(\mathcal{A})$): $0 \to X \hookrightarrow I^0 \to I^1 \to \cdots$.